Pulsar Candidate Classification with Machine Learning

Capstone Project (AAI-590-01/02)

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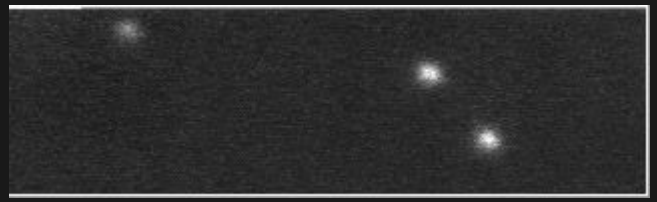
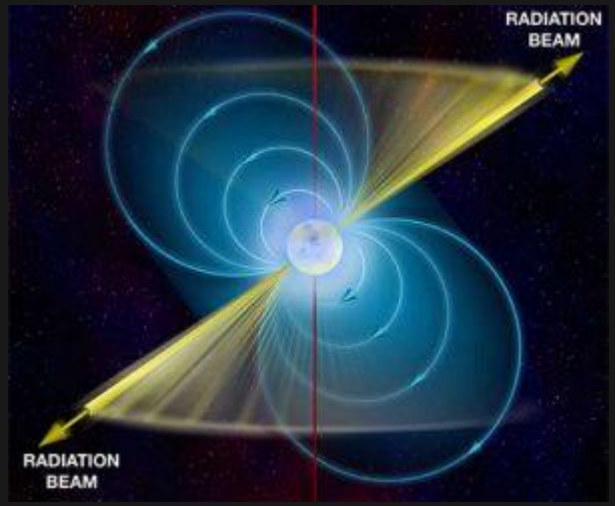
Scientific Background



- Pulsars are rotating neutron stars emitting periodic radio signals.
- Used in detecting gravitational waves and testing relativity.
- Some types of pulsars are more accurate than atomic clocks.
- Modern surveys generate millions of candidates. Most signals are noise, manual filtering is inefficient.

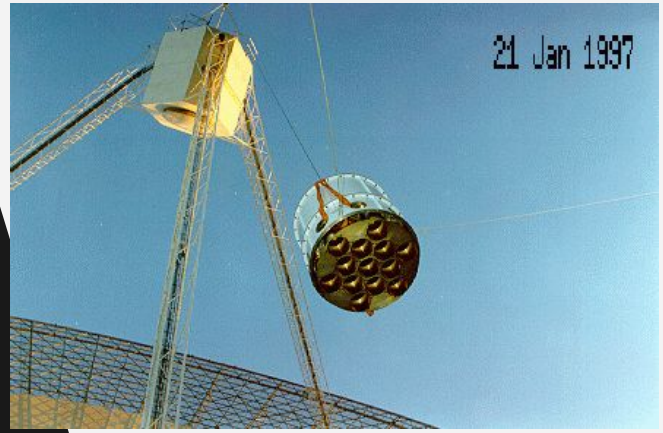


Parkes Observatory, Australia

HTRU2: High Time Resolution Universe Survey (UCI)

The radio telescope uses multiple beams to scan several regions simultaneously.

The pipeline tests signal delays and identifies where the signal is strongest. The shape of this signal (mean, skewness, kurtosis) is summarized and recorded.



Data Source

- HTRU2 dataset (17,898 samples)
- 8 numerical features per signal.
- Signal shape (profile) and dispersion curve (DM): dispersion corrects varios delays by frequency.

- Imbalanced data (<10% pulsars)
- High false positives from noise

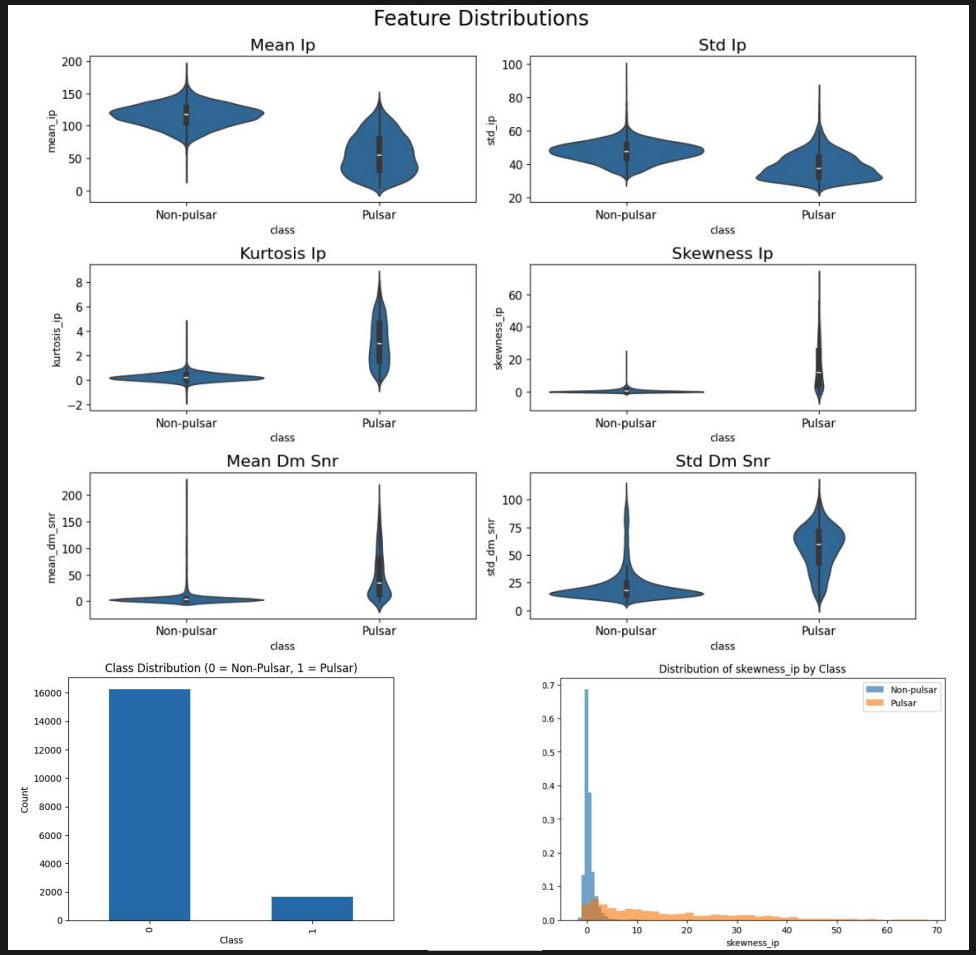
- Need accurate and scalable classification
- Maximize detection but min. false positives

Variables Table

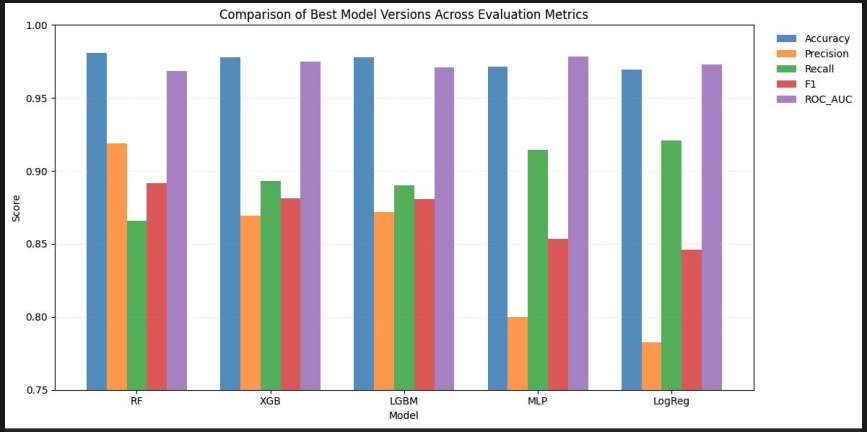
Variable Name	Role	Type
Profile_mean	Feature	Continuous
Profile_stdev	Feature	Continuous
Profile_skewness	Feature	Continuous
Profile_kurtosis	Feature	Continuous
DM_mean	Feature	Continuous
DM_stdev	Feature	Continuous
DM_skewness	Feature	Continuous
DM_kurtosis	Feature	Continuous
class	Target	Binary

Exploratory Data Analysis

- Class weighting for imbalance.
- Higher-order statistical features (skewness, kurtosis) are most informative.
- Nonlinear patterns suggest tree-based models can be suitable.



Model Comparison



Best F1: Random Forest
Best macro metric: XGBoost
Best recall: Logistic Regression / MLP

Table 5.1: Comparison of Models' Performance

Model	Acc.	Prec.	Recall	F1	ROC –AUC
RF	**0.980**	**0.919**	**0.866**	**0.892**	**0.969**
XGB	0.978	0.869	0.893	0.881	0.975
LGBM	0.978	0.872	0.890	0.881	0.971
MLP	0.971	0.800	0.915	0.854	0.979
LR	0.969	0.782	0.921	0.846	0.973

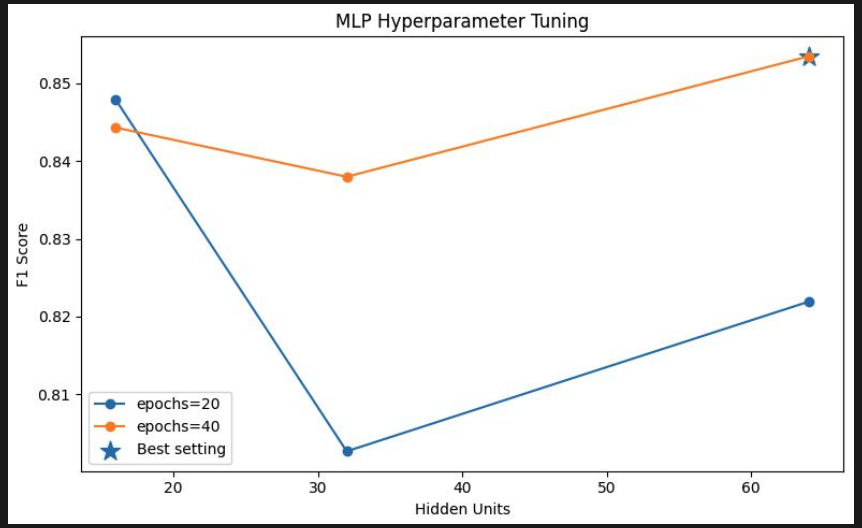
Observation: Ensemble models capture nonlinear feature interactions better than linear/MLP

MLP

Increasing model complexity led to only very small improvements.

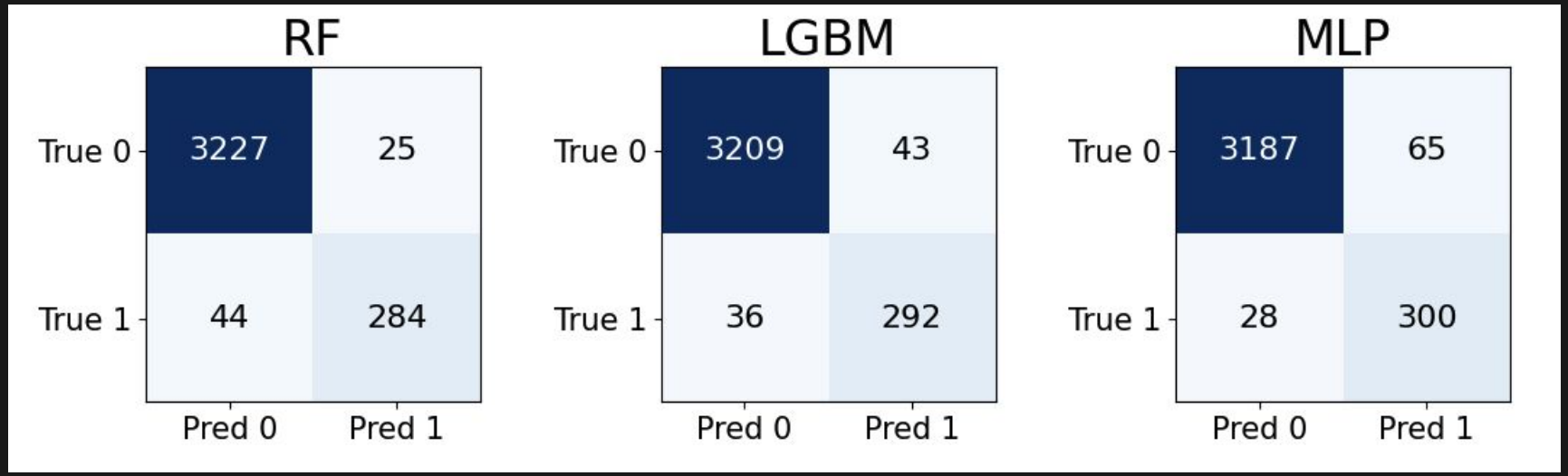
The limited feature space restricts the benefit of added complexity and can lead to overfitting.

Neural networks are not ideal for this type of structured tabular data.



Model Comparison

Random Forest achieves highest F1 (0.8917)
XGBoost improves recall (better rare signal detection)



Results and Conclusion

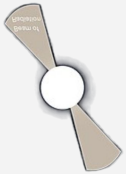
- **Best Overall**: Random Forest
  F1 = 0.8917

- Tree-based models perform best on this dataset.

- However, best choice also depends on application goals:

  High precision allows for fewer false detections , but high recall gives better pulsar discovery).

Future Direction

- Real-time pipeline: signal → features → model scoring
- Move toward raw signal representations (time-series inputs)
- Hybrid approach: high-recall + high-precision models
- Rank candidates by pulsar probability

Thank you!